



What building parallax established, and what P01 must change

— 144 tests · 28 commits at the whole-branch review — the state described here

Tool: Task-force-for-AI-agents-in-Healthcare/parallax

This records what the implementation settled, what it forced us to correct, and what P01 needs to say differently as a result. It exists because the working notes lived in git-ignored scratch, and these conclusions should not.

This is a snapshot of the trust calculus alone. The counts on the cover are the repository as it stood at the twelve-task whole-branch review, and they are correct for it; they are not current. Work that landed afterwards — real Intel TDX quote verification, RA-TLS key binding, collateral fetch and cache, and `parallax-proxy` — is not described here and did not exist when this was written. For what the tool does now, and for current figures, read the repository `README.md` and `docs/STANDARD-MAP.md`. Nothing below has been revised in the light of that work.

Amendments P01 needs

1. The termination argument is simpler than the paper claims, and now has evidence.

P01's mechanization section proposes SLG resolution with tabling, in the style of XSB, to stop circular attestation from diverging. That is not what the tool does and not what is needed. Modelling the residual trust set as a bounded lattice — join is set union over a finite principal set, so the powerset has finite height — makes the least fixpoint terminate by construction, regardless of cycles in the delegation graph. Termination falls out of the datatype rather than the resolution strategy.

This has been exercised well past the point of doubt: a 300-node delegation cycle spanning two mechanism layers terminates in about 0.01s, and a 40-node complete delegation graph with all 1,600 edges including self-loops terminates in about 1 ms. **Rewrite the mechanization section around the bounded-lattice argument** and drop the tabling proposal; it is a weaker claim requiring more machinery.

Note also that only *one* of the two lattices carries termination. The latency lattice has infinite height ($\text{Bounded}(0) < \text{Bounded}(1) < \dots$); the fixpoint never iterates in it. Saying “two lattices of finite height” is false, and a formal-methods referee catches it immediately.

2. §4's independent-encoding experiment needs an order-independence claim, and the reason is not academic.

P01 offers independent double-encoding as its answer to the completeness problem: two authors encode the same deployment and any divergence is a finding about the calculus. That experiment is worthless unless the comparison is invariant to how the deployment was written down, and during development it twice was not.

- Mechanism identity was initially *positional* (`tee_attestation#0`), so a deployment compared as INCOMPARABLE to a re-ordering of itself. All twelve non-identical example pairs returned INCOMPARABLE, meaning the acceptance test carrying the paper's headline claim would have passed against an implementation that returned INCOMPARABLE unconditionally.

- After that was fixed, identity was still *lexical* in duration fields, so `collateral_refresh = "12h"` and `"720m"` — identical in every computed value — again produced total divergence.

Both are fixed and pinned by tests. The paper should state plainly that the comparison is order-independent and duration-normalised **by construction**, because two independent encoders writing 12h and 720m is not a hypothetical.

3. **Do not overstate the incomparability result.** The tool ranks only pairs that establish the same claim; comparing trust sets for different propositions is refused. Of the twelve ordered pairs across the four non-hybrid examples, eight are N/A on that ground and four are INCOMPARABLE — that is two unordered pairs, Σ_1/Σ_3 and Σ_2/Σ_4 . The claim is real but narrower than “trust sets come out incomparable” implies.
4. **The evidence base is five self-authored deployments and no real-world system.** The tool reproduces the hand-derived five-party TDX set and finds the shared build-pipeline dependency in the hybrid, with a negative control that correctly reports nothing. That is the whole of it. Say so.

What the implementation confirmed

- **The five-party TDX set is derivable from the deployment description**, not hardcoded — verified by collapsing a principal in the input and watching the count track it. Exactly one of the five, the collateral authority, has a bounded detection latency; the other four are silent and permanent. That asymmetry is the paper’s central empirical claim and it holds.
- Σ_2 and Σ_4 **are genuinely incomparable** — five disjoint assumptions on each side, same claim, no artifact of the encoding.
- **The hybrid’s two layers do share a build pipeline**, and the independent-pipelines variant reports nothing. The finding is discovery, not theatre.
- **Anchoring is a trade, not a pure gain.** Anchoring to a settlement layer converts log-operator trust into a bounded detection window and silently ingests a consensus-execution-fidelity assumption. Both appear in the manifest.

The defect pattern worth carrying to transit and occultation

Every Important-or-worse finding across twelve tasks and the final review was a defect in the **plan**, not in an implementation. Five were Critical, and all five shared a shape: **correct code implementing a subtly wrong specification, producing a confident wrong answer rather than an error.**

Defect	Wrong answer it produced
Positional mechanism tag	a file INCOMPARABLE to a reordering of itself
Lexical duration identity	12h vs 720m reported as total divergence
Flat "delegation" tag	a single-layer deployment reporting shared dependencies
via_delegation per principal	a dead-end delegate named as a reason a principal matters
Policy without deny_unknown_fields	a one-letter typo silently disabling the gate; check exits 0

Four of the five trace to one root cause: **a field participating in a value's identity was specified casually**. Anything that feeds `Eq/Ord/Hash` is load-bearing for every comparison downstream, and specifying it loosely produces answers that look right.

The two checks that would have caught all five, and which belong in any plan of this kind:

1. **For every derived identity, test that semantically identical inputs compare equal** — reordered, respelled, re-serialised.
2. **For every configuration surface, reject unknown keys**. A silently ignored typo in a security policy is a gate that fails open.

The reviewers earned their cost by going to primary sources rather than trusting the plan's paraphrase — reading `ascent_base`'s `Lattice` trait instead of the brief's summary of it, rebuilding pre-fix binaries to confirm a repro, and mutation-testing claims that tests were supposed to pin. That last technique found the delegation-direction gap that five passing tests missed.

Carried, not fixed

None affects a claim the paper makes.

- Table 2 in the paper is pinned to the code by a tripwire test covering mechanism, capability, latency and impact — but nothing pins the caption's prose to the table. Three separate factual errors have been found in that caption; generating the table from source is the real fix.
- The CI paper job has never run; `setup-tectonic` on Ubuntu is unverified.
- `shared_dependencies(d, t)` requires `t == solve(d)` by convention, not by type. A `Solved` newtype would remove this and `explain`'s vestigial parameter together.
- `policy::evaluate` does not itself verify `$schema`; the CLI does.